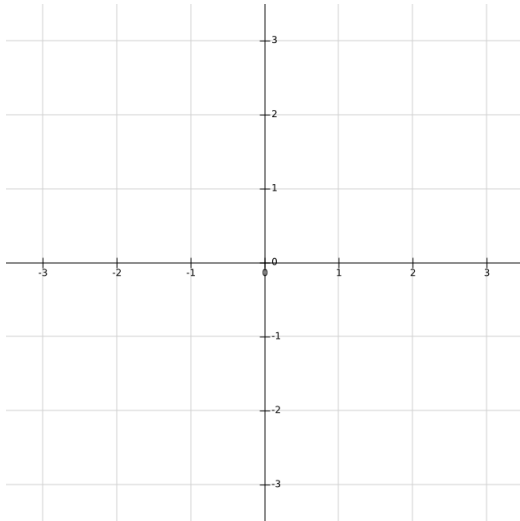


COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2020 – Paper 7

7 Further Graphics (pb355)

Voronoi Diagrams (a) Consider the set $\mathbf{C}$ of 2D control points:
 $\mathbf{C} = \{ (0, 0), (0, 2), (2, 1), (2, -1), (-2, -1), (-2, 1) \}$



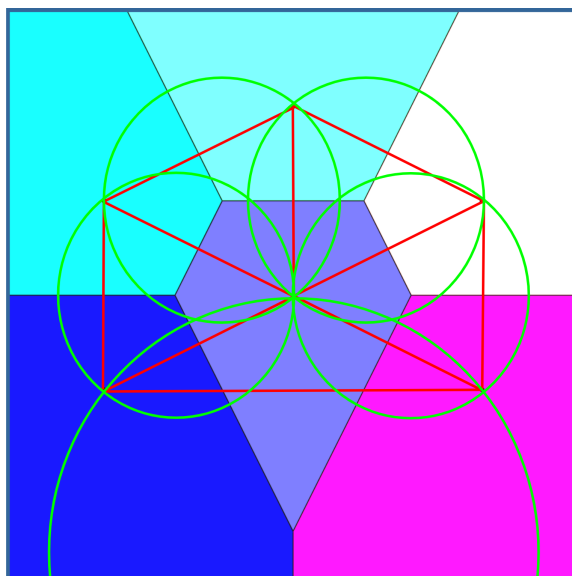
On 3 separate 2D graph plots, each ranging from $(-3, -3)$ to $(3, 3)$,

(i) Draw the *Voronoi diagram* of $\mathbf{C}$ [2 marks]

(ii) Draw the *Delaunay triangulation* of $\mathbf{C}$ [2 marks]

(iii) Draw the *empty circles* of the *Voronoi points* of $\mathbf{C}$ [2 marks]

Answer:



- The six cells are the Voronoi diagram
- The red edges are the Delaunay triangulation

— *Solution notes* —

- The green circles are the empty circles

Marking scheme: 2 points each for the correct answer or 1 point for a wrong answer that shows an understanding of the definition of the relevant term

- (iv) What is the first value in the *equiangularity* of **C**? [3 marks]

Answer:

An equiangularity is the ascending sorted list of the angles of the triangles of a triangulation. By inspection of the figure above, the least angle (the most acute) is the angle of the two side vertices of the bottommost triangle. Solve for the angle between line segments $(-2, -1) \rightarrow (2, -1)$ and $(-2, -1) \rightarrow (0, 0)$:

$$\theta = \arccos \left(\frac{(4,0)}{|(4,0)|} \cdot \frac{(2,1)}{|(2,1)|} \right) = \arccos \left(\frac{2}{\sqrt{5}} \right) = 0.463647609 \text{ radians}$$

Marking scheme: full points for the right answer, or,

- 1 point for showing an understanding of the definition of the equiangularity
 - 1 point for trying to do the right thing but getting the maths wrong
-

- (v) What is the (X, Y) position of the Voronoi point of **C** with the most negative Y coordinate? [3 marks]

Answer:

Find the Y -intercept of the line perpendicular to the line between $(0, 0)$ and $(2, -1)$ which passes through the point $(1, -\frac{1}{2})$ equidistant from both:

Rotate vector $(1, -\frac{1}{2})$ 90 degrees: $(-\frac{1}{2}, -1)$

$$Y_0 = Y + \frac{-X}{dx} dy = -\frac{1}{2} + -\frac{-1}{-\frac{1}{2}} = -2\frac{1}{2}$$

$\rightarrow$ the Voronoi point is at $(0, -2\frac{1}{2})$

Marking scheme: full points for the right answer, or,

- 1 point for showing an understanding of the definition of the Voronoi point
 - 1 point for trying to do the right thing but getting the maths wrong
-

- Voronoi Diagrams (b) Using pseudocode, give an algorithm for finding the Delaunay triangulation of a set of 2D points **S**. [4 marks]

Answer:

Several algorithms were identified in Lecture 5; any are suitable here.

Marking scheme: for full marks, the pseudocode given must be of sufficient quality and detail that a novice could feasibly implement the algorithm described

- Voronoi Diagrams (c) Explain why the empty circles around the Voronoi points of a Voronoi diagram are, in fact, empty. [4 marks]

Answer:

The radius of the empty circles around each Voronoi point is the distance to the three or more nearest control points. As they're nearest, no other control point can be nearer. So by definition the circles must be empty of other control points.

Marking scheme: full points for a clear and simple argument or proof. Points will be deducted if the argument is illegible or incoherent.
